

Inverse Kinematics

Using Trigonometry

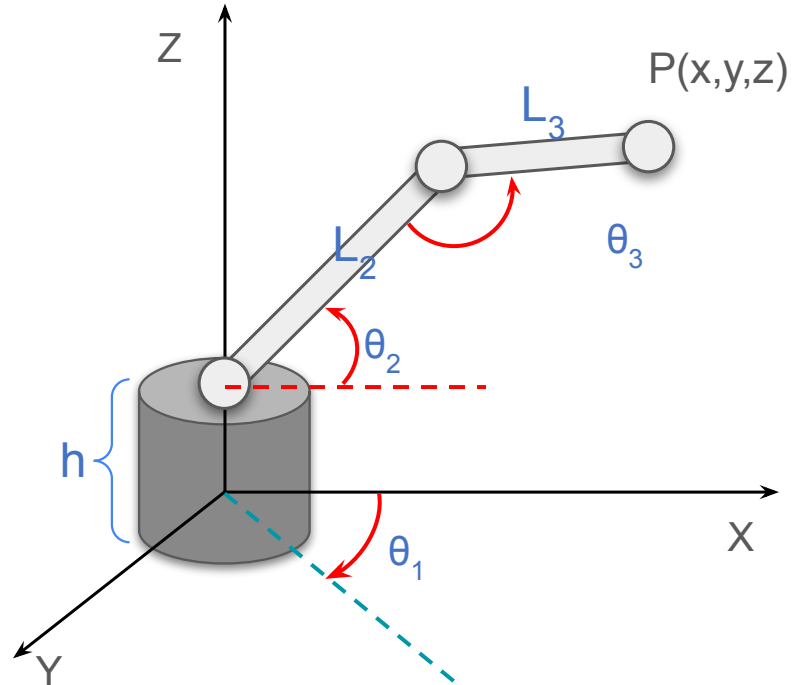
Inverse Kinematics using Trigonometry

The figure illustrates a 3R-3D Planar Robot Manipulator capable of reaching any point $P(x,y,z)$ in a XYZ-workspace

In inverse kinematics, the **goal** is to determine the **joint angles** $(\theta_1, \theta_2, \theta_3)$ for a given end-effector position $P(x,y,z)$

This problem cannot be solved using a purely 2D geometric approach. Therefore, the manipulator is conceptually **decomposed into two view ports**, similar to the forward kinematics formulation:

- Top View - captures rotational motion
- Side View - captures movement of other links



Top View

For inverse kinematics, we need to solve for the top view first

From the top view, we simply see how the arm is rotated around the base and how the projected length **d** spreads into the X and Y directions.

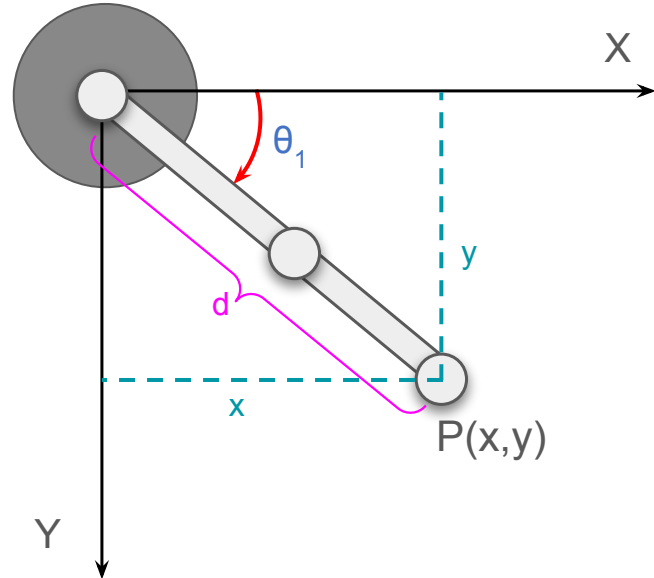
Considering **P(x, y)** are given, we get,

$$\tan\theta_1 = y / x$$

$$\therefore \theta_1 = \tan^{-1}(y / x)$$

For further calculations in the side view, we need to calculate the value of d.

$$d = \sqrt{(x^2 + y^2)}$$



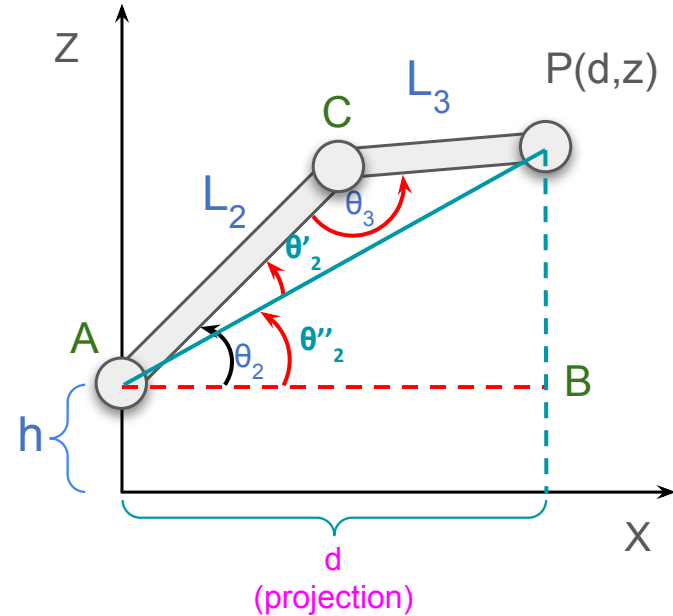
Side View

In side view, d (the projection of the arm) becomes the value of x .

In the modified side-view formation, θ_2 is divided into two parts.

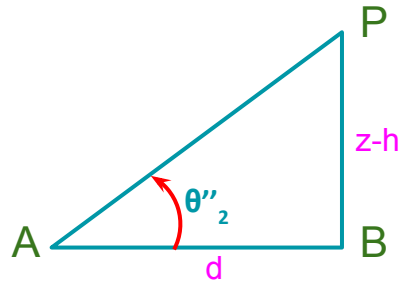
$$\theta_2 = \angle CAP (\theta'_2) + \angle PAB (\theta''_2)$$

Now we need to separately calculate θ'_2 and θ''_2



Side View

From $\triangle APB$, we get



$$\theta''_2 = \tan^{-1}\left(\frac{z-h}{d}\right)$$

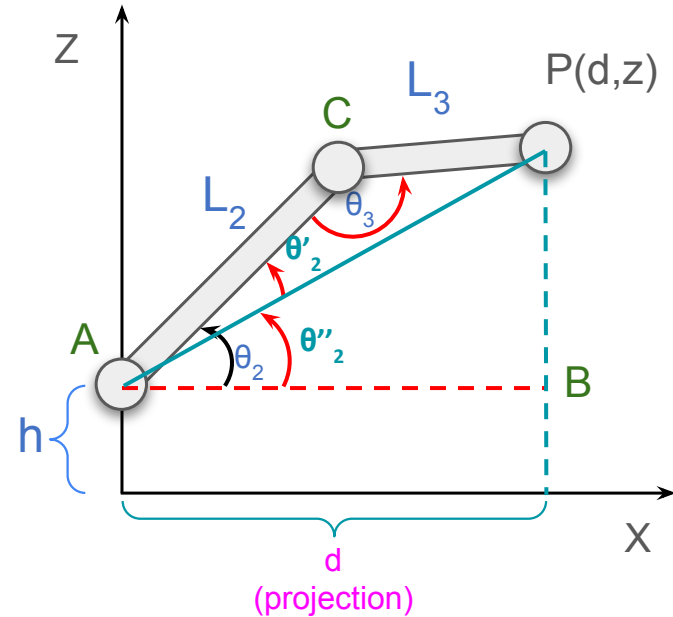
$$\theta''_2 = \tan^{-1}\left(\frac{z-h}{\sqrt{(x^2 + y^2)}}\right)$$

and,

$$AP^2 = d^2 + (z-h)^2$$

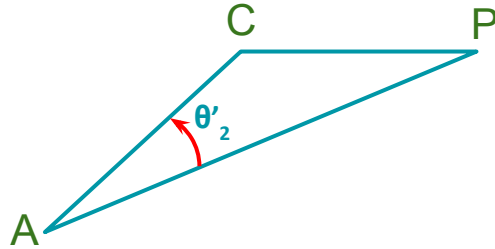
$$AP^2 = (\sqrt{(x^2 + y^2)})^2 + (z-h)^2$$

$$AP = \sqrt{(x^2 + y^2 + (z-h)^2)}$$



Side View

From $\triangle CAP$, we get



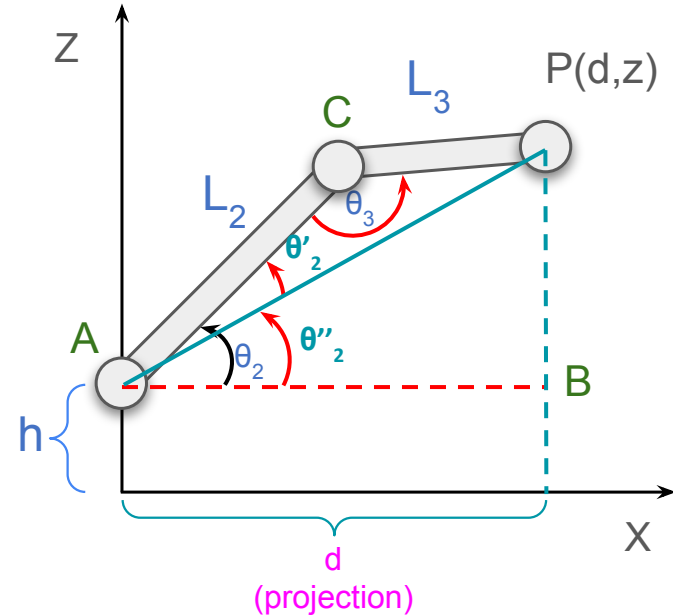
$$\cos \theta'_2 = \frac{AC^2 + AP^2 - CP^2}{2 \cdot AC \cdot AP} \quad (\text{cosine rule})$$

$$\theta'_2 = \cos^{-1} \frac{L_2^2 + AP^2 - L_3^2}{2 \cdot L_2 \cdot AP}$$

Finally,

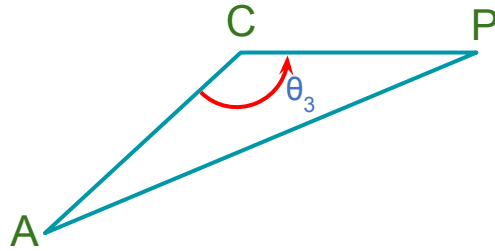
$$\theta_2 = \theta'_2 + \theta''_2$$

$$\theta_2 = \cos^{-1} \left(\frac{L_2^2 + AP^2 - L_3^2}{2 \cdot L_2 \cdot AP} \right) + \tan^{-1} \left(\frac{z - h}{\sqrt{(x^2 + y^2)}} \right)$$



Side View

From $\triangle CAP$, we also get



$$\cos \theta_3 = \frac{AC^2 + CP^2 - AP^2}{2 \cdot AC \cdot CP} \quad (\text{cosine rule})$$

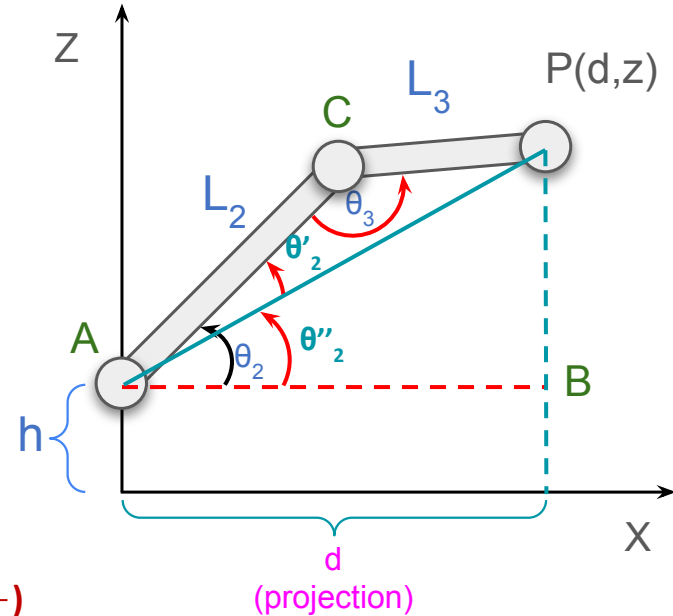
$$\theta_3 = \cos^{-1} \frac{L_2^2 + L_3^2 - AP^2}{2 \cdot L_2 \cdot L_3}$$

Results:

$$\theta_1 = \tan^{-1}(y / x)$$

$$\theta_2 = \cos^{-1} \left(\frac{L_2^2 + AP^2 - L_3^2}{2 \cdot L_2 \cdot AP} \right) + \tan^{-1} \left(\frac{z - h}{\sqrt{x^2 + y^2}} \right)$$

$$\theta_3 = \cos^{-1} \frac{L_2^2 + L_3^2 - AP^2}{2 \cdot L_2 \cdot L_3}$$



Inverse Kinematics using Trigonometry [Extension]

$$\theta_1 = \tan^{-1}(y / x) \text{ or, } (90 - \tan^{-1}(y / x))$$

$$\theta_2 = \cos^{-1}\left(\frac{L_2^2 + AP^2 - L_3^2}{2 \cdot L_2 \cdot AP}\right) + \tan^{-1}\left(\frac{z - h}{\sqrt{x^2 + y^2}}\right)$$

$$\theta_3 = 180 - \cos^{-1}\frac{L_2^2 + L_3^2 - AP^2}{2 \cdot L_2 \cdot L_3}$$

Here,

$$AP^2 = (\sqrt{x^2 + y^2})^2 + (z - h)^2$$

